

Hossam Eid

AI/ML ENGINEER
DATA SCIENTIST

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Kazan, Russia · remote / hybrid · open to relocation · RF residence permit,
unrestricted work

AI/ML engineer building end-to-end ML solutions — from data preparation and model training to backend services, APIs and Docker deployment. Focus areas: predictive analytics and time-series analysis, LLM/RAG systems and computer vision. Master's student at ITIS, Kazan Federal University (Artificial Intelligence in Digital Product Development); completed my bachelor's with honours. My thesis — accuracy 0.97 in fault diagnosis and R^2 0.86 in remaining-useful-life prediction — passed technical review by JSC TVEL (Rosatom State Corporation).

EXPERIENCE

AI/ML Engineer

January 2024 — present

Kazan Federal University, ITIS · Kazan · artificial intelligence & machine learning projects

Key project — remaining useful life prediction & fault diagnosis for power transformers (bachelor's thesis, 2025)

- Time-series ML pipeline: feature extraction with tsfresh and TSFEL, removal of correlated features, scaling, SHAP feature attribution and a FEDOT AutoML baseline for comparison; dataset of 3,000 series (H_2 , CO, C_2H_4 , C_2H_2).
- Fault diagnosis (4-class classification): stacking ensemble of Random Forest, SVM and XGBoost with a logistic-regression meta-model — **accuracy 0.97, macro-F1 0.92**.
- Remaining useful life (regression): compared 11 models including LightGBM, CatBoost and recurrent networks (GRU / LSTM), best LightGBM — **R^2 0.86, MAE 66.3, RMSE 91.83**.
- Microservice architecture: Django REST Framework, a dedicated Flask ML-serving API, SQLite3, JWT, WebSockets (Django Channels). Frontend: React + TypeScript + Vite, React Query and Zustand, Tailwind CSS, Plotly.js charts, PDF export (jsPDF), multilingual UI (i18next).
- Automated PDF reporting, e-mail notifications, a domain chat assistant on a local Qwen 2.5 (Q&A over diagnostic results with no external services); Docker deployment. Repository: github.com/hossameid7/graduation_project
- Outcome: the work passed technical review by JSC TVEL (Rosatom) and was forwarded for implementation review.

Mentoring & technical consulting

- Assist my academic supervisor with the technical supervision and review of **45+ bachelor ML/AI theses**: problem framing, metric selection, solution architecture (data → features → model → service) and code review.
- Diagnose common mistakes — data leakage, incorrect validation, class imbalance — and prepare projects for defence. Hands-on mentoring and code-review experience across two graduating years (2024/25, 2025/26).
- Domains: LLM/RAG and AI agents, computer vision in medicine (MRI, X-ray) and industry, time-series analysis (RUL prediction, energy, fintech), graph neural networks, NLP.

PROJECTS

LLM Guardrails Safety Gateway — security proxy for LLM applications

2026

Asynchronous gateway between the application and the LLM: PII masking (e-mails, cards, phone numbers, SSNs) with restoration in the response, prompt-injection and jailbreak blocking through 35+ rules and structural heuristics, a ChromaDB + MiniLM semantic cache (0.92 threshold, < 10 ms hits), plus p50/p95/p99 latency, token and security-violation telemetry.

Python · FastAPI · LangChain · Llama 3.3 70B (Groq) · ChromaDB · Sentence-Transformers · Pydantic v2 · Docker · github.com/hossameid7/LLM-Guardrails-Safety-Gateway

Multi-Agent Log Analyzer — log analysis & automated incident remediation

2026

Three agents in a LangGraph state graph with typed shared state (log parser → RAG retrieval → fix generator). Returns a Pydantic-validated report: root cause, numbered steps with Bash commands, preventive measures. FastAPI + Telegram bot; the LLM backend switches between OpenAI-compatible providers and a local Ollama; 27+ pytest tests.

Python · LangGraph · LangChain · ChromaDB · FastAPI · Telegram Bot API · pytest · Docker · github.com/hossameid7/multi-agent-log-analyzer

Medical AI: tumour classification & segmentation

2026

VGG16-based classification (accuracy 87%, F1 0.87) and a U-Net trained from scratch with batch normalisation (val. accuracy 94.7%) in a single pipeline; Streamlit demo and a Flutter mobile interface. Research prototype, not a medical device.

Python · TensorFlow / Keras · VGG16 · U-Net · Streamlit · Flutter · github.com/hossameid7/flutter-streamlit-breast-ai

On-device brain tumour detection from MRI2026

Binary classification (Xception — test accuracy 98.02%, validation 97.00%) and semantic segmentation (U-Net with a VGG16 backbone — val. accuracy 98.46%); inference runs entirely on-device through native C++ modules, so latency drops and scans never leave the phone. Research prototype, not a medical device.

Python · TensorFlow / Keras · Xception · U-Net (VGG16) · C++ · CMake · Flutter ·
github.com/hossameid7/flutter-brain-tumor-ai

NLP assistant for legal translation & PDF analysis2026

Segments legal PDFs into semantic blocks for context-aware translation, side-by-side comparison of source and translation, automatic glossary extraction and SVG stamp generation.

Python · LLM · LangChain · NLP · PDF parsing ·
github.com/hossameid7/AI-Legal-Translation-Assistant

Telegram AI-assistant builder · Consent Manager (Russian data-protection law)2025 – 2026

A builder for LLM + RAG Telegram assistants with user-supplied knowledge bases and technical-documentation generation. Consent Manager — a service for managing personal-data consents under Russian law 152-FZ: issuing and revoking consent by e-mail/PIN, PDF documents, admin panel.

Python · Hugging Face · LangChain · FAISS · LoRA / QLoRA · Flask · SQLAlchemy · Telegram Bot API

Full project list:
github.com/hossameid7

EDUCATION

Master's — Software Engineering (09.04.04)2025 – 2027

Kazan Federal University, ITIS · Kazan · track: Artificial Intelligence in Digital Product Development

Bachelor's — Software Engineering (09.03.04)2021 – 2025

Kazan Federal University, ITIS · Kazan · track: Information Systems Development Technologies ·
graduated with honours (red diploma)

Professional retraining — Fundamentals of Data Platform Development2024 – 2025

Kazan Federal University · Kazan · 252 hours · qualification: Programmer

Preparatory faculty — Russian language (science track)2021

Kazan State Academy of Veterinary Medicine · Kazan

SKILLS

ML / DATA SCIENCEPython · SQL · pandas · NumPy · scikit-learn · PyTorch · TensorFlow / Keras · RNN / GRU / LSTM · LightGBM · XGBoost · CatBoost · stacking ensembles · cross-validation · feature engineering (tsfresh, TSFEL) · SHAP · AutoML (FEDOT) · feature selection & scaling · class imbalance · time series · anomaly detection · EDA · leakage-free validation

LLM / NLP
RAG · LangChain · LangGraph (multi-agent graphs) · Hugging Face Transformers · LoRA / QLoRA · vector search (ChromaDB, FAISS) · Sentence-Transformers embeddings · semantic caching · structured output (Pydantic) · prompt engineering · prompt-injection defence · PII masking · Groq / OpenAI-compatible APIs · Ollama

COMPUTER VISION
OpenCV · CNN · transfer learning (VGG16, Xception) · U-Net · batch normalisation · data augmentation · image preprocessing · segmentation & classification · on-device inference (C++)

BACKEND / MLOPS
FastAPI (async) · Flask · Django REST Framework · REST API design · JWT · WebSockets · Pydantic v2 · PostgreSQL · SQLite · SQLAlchemy · Docker & docker-compose · pytest · Git · Linux · Streamlit · React / TypeScript / Vite · React Query · Zustand · Tailwind CSS · Plotly.js · Telegram Bot API

PRACTICE
Reproducible experiments · honest metrics and baselines · diagnosing data leakage and invalid validation · automated tests · logging and latency metrics · containerisation and deployment · technical documentation

LANGUAGES
Arabic — native · English — C2, proficient · Russian — C1, advanced

AWARDS & CERTIFICATES

— Prize winner, Open Doors: Russian Scholarship Project — applied mathematics & AI track · 2025

— Letters of appreciation, FC Rubin — organising the Kubok Vostoka tournament · 2025

— Top international graduate, ITIS, Kazan Federal University · 2025

— Letter of appreciation from the KFU vice-rector for contribution to student self-governance · 2025

— All-Russian Mathematics Dictation — 15/15, T-Education · 2025

— Semifinalist, Volga-IT'24 olympiad — artificial intelligence & data analysis track · 2024

- Speaker Certificate — Sharjah International Conference on AI & Linguistics (SICAL), UAE · 2024
- Certificate: consultant in developing digital literacy (digital curator) · 2024
- Finalist certificate, KFU Student of the Year · 2024
- Letter of appreciation from the ITIS KFU director for contribution to student life · 2022
- Letters of appreciation from the ITIS KFU director for mentoring (tutoring) · 2022